Guillermo Serrera

C. Santa Lucía 43, 39003 Santander, Cantabria, Spain
gserrera@gmail.com
ORCID ID: 0000-0001-9363-187X
Google Scholar profile

Research profile

Physicist specializing in nanophotonics and light-matter interactions, with research experience in high-index dielectric nanoantennas and their application to enhanced chiral sensing. Author of 9 peer-reviewed journal articles, including 5 first-author publications, with international research experience at Ghent University (Belgium) and the University of Fribourg (Switzerland). Experienced in computational nano-optics techniques, scientific programming and polarization-resolved Raman microscopy.

Research keywords

Nanophotonics · Dielectric nanoantennas and metasurfaces · Chiral nano-optics · Plasmonics · Computational nano-optics · Stokes polarimetry

Research metrics

9 peer-reviewed publications (5 as first-author).
Total number of citations: 101 (Google Scholar).
h-index: 5 (Google Scholar).
10 oral conference contributions as presenting author (3 invited talks)
5 poster contributions

Employment

2023–26 **Predoctoral researcher**, University of Cantabria

Education

University education

2022–26* **Ph.D.** in Science and Technology, University of Cantabria
2021–22 **M.Sc.** in Science and Engineering of Light, University of Cantabria
2016–21 **B.Sc.** in Physics, University of Cantabria

Other training

2025 ACS Reviewer Lab (2 hours), American Chemical Society.
2025 Optica Publishing Group Certified Reviewer Course (1 hour), Optica Society of America.
2025 Fundamentals and applications of optical materials (20 hours), University of Cantabria summer school.
2024 Certified Peer Reviewer Course (4 hours), Elsevier Researcher Academy.

*Expected.

2022 European School on Plasmonics and Phase Change Materials (16 hours), PHEMTRONICS H2020 FET Open project.

Research Experience

2023–26 Researcher in the project "MOdelling of the PHOtothermal response of hybrid nanoSYStems (MOPHOSYS)" Advisor: Pablo Albella (University of Cantabria), Spain.

2021–22 Part time researcher in the project "Spectroscopy of planetary and atmospheric particulae by optical tweezers". Advisor: Onofrio Maragò (IPCF-CNR), Italy.

Awards, Funding & Honors

2025 Funding for short research stays by the Spanish Ministry of Science, Innovation and Universities (EST25/00316).

2025 Best theoretical poster at CEN2025 conference.

2023 Best poster at University of Cantabria Science day.

2023–26 University Professorship Training (FPU21/02296) Fellow at the University of Cantabria. Funded by the Spanish Ministry of Science, Innovation and Universities.

2020–21 Collaboration Research Grant at the Applied Physics Department of the University of Cantabria. Funded by the Spanish Ministry of Science, Innovation and Universities.

Publications

G Google Scholar

Journal Articles

J1. **Guillermo Serrera**, Gutiérrez, Y., Gil, J. J. & Moreno, F. "Stokes-Based 3D Polarimetric Analysis of Dipolar Near-Fields". *Advanced Photonics Research* **7,** e70251. https://doi.org/10.1002/adpr.70251 (2026).

J2. **Guillermo Serrera** & Albella, P. "Enhanced Enantioselective Optical Trapping enabled by Longitudinal Mie Resonances in Silicon Nanodisks". *Nanophotonics* **15,** e70207. https://doi.org/10.1002/nap2.70207 (2026).

J3. Sanz-Paz, M., Siegel, N., **Guillermo Serrera**, González-Colsa, J., Zhu, F., Kołątaj, K., Fujii, M., Sugimoto, H., Albella, P. & Acuna, G. P. "Color Routing and Beam Steering of Single-Molecule Emission with a Spherical Silicon Nanoantenna". *Advanced Functional Materials* **36,** e29955. https://doi.org/10.1002/adfm.202529955 (2026).

J4. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. "3-dimensional plasmonic nanomotors enabled by independent integration of optical pulling and lateral forces". *Nanophotonics* **14,** 2500299. https://doi.org/10.1515/nanoph-2025-0374 (2025).

J5. Siegel, N., Sanz-Paz, M., González-Colsa, J., **Guillermo Serrera**, Zhu, F., Szalai, A. M., Kołątaj, K., Fujii, M., Sugimoto, H., Albella, P. & Acuna, G. P. "Distance-Dependent Interaction between a Single Emitter and a Single Dielectric Nanoparticle Using DNA Origami". *Small Structures* **6,** 2500299. https://doi.org/10.1002/sstr.202500299 (2025).

J6. **Guillermo Serrera**, González-Colsa, J. & Albella, P. "Amplified linear and nonlinear chiral sensing assisted by anapole modes in hybrid metasurfaces". *Applied Physics Letters* **124,** 251701. https://doi.org/10.1063/5.0212393 (2024).

J7. **Guillermo Serrera**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. "Enhanced optical chirality with directional emission of Surface Plasmon Polaritons for chiral sensing applications". *Journal of Quantitative Spectroscopy and Radiative Transfer* **284,** 108166. https://doi.org/10.1016/j.jqsrt.2022.108166 (2022).

J8. González-Colsa, J., **Guillermo Serrera**, Saiz, J. M., Ortiz, D., González, F., Bresme, F., Moreno, F. & Albella, P. "Gold nanodoughnut as an outstanding nanoheater for photothermal applications". *Optics Express* **30,** 125–137. https://doi.org/10.1364/OE.446637 (2022).

J9. González-Colsa, J., **Guillermo Serrera**, Saiz, J. M., González, F., Moreno, F. & Albella, P. "On the performance of a tunable grating-based high sensitivity unidirectional plasmonic sensor". *Optics Express* **29,** 13733–13745. https://doi.org/10.1364/OE.422026 (2021).

Conference contributions (as presenting author)

Talks

T1. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Light-Driven Nanomotors with controllable 3D Motion.* SPIE Optics + Photonics 2026 (San Diego, United States). Aug. 2026.

T2. **Guillermo Serrera**, Godiksen, R. H., Dedding, L. M., Sistermans, T. T. C., Mohammadi, E., Raziman, T. V., S., R., Albella, P. & Curto, A. G. *Circularly polarized Raman scattering for optical spin detection in silicon metasurfaces.* Invited talk at META 2026 (Dublin, Ireland). July 2026.

T3. **Guillermo Serrera**, Godiksen, R. H., Dedding, L. M., Sistermans, T. T. C., Mohammadi, E., Raziman, T. V., S., R., Albella, P. & Curto, A. G. *Probing optical spin with circularly polarized Raman scattering in silicon metasurfaces.* Invited talk at 14th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2026) (Marbella, Spain). Mar. 2026.

T4. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Control of 3-Dimensional Motion in Plasmonic Nanomotors with Optical Pulling Forces.* Invited talk at META 2025 (Málaga, Spain). July 2025.

T5. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Optical pulling forces for 3-dimensional motion control in plasmonic nanomotors.* 13th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2025) (Porto, Portugal). Feb. 2025.

T6. **Guillermo Serrera**, González-Colsa, J. & Albella, P. *Anapole modes enabling linear and nonlinear chiral sensing in hybrid metasurfaces.* META 2024 (Toyama, Japan). July 2024.

T7. **Guillermo Serrera**, González-Colsa, J. & Albella, P. *Hybrid metal-dielectric resonator for anapole-assisted enhanced chiral sensingapplications.* SPIE Photonics Europe (Strasbourg, France). Apr. 2024.

T8. **Guillermo Serrera**, González-Colsa, J. & Albella, P. *Exploiting Hybrid Metasurfaces for enhanced Linear and Nonlinear Chiral Sensing Applications.* 12th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2024) (Rome, Italy). Feb. 2024.

T9. **Guillermo Serrera**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. *Chiral sensing based on directional plasmons.* International Conference of Quantum, Nonlinear and Nanophotonics 2022 (ICQNN'22) (Jena, Germany). Sept. 2022.

T10. **Guillermo Serrera**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. *Directional Plasmon Emission in a Mixed Dielectric-plasmonic Structure: Application to Chiral Sensing.* 10th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2022) (Online). Feb. 2022.

Posters

P1. **Guillermo Serrera**, Gutiérrez, Y., Gil, J. J. & Moreno, F. *3D Stokes Polarimetry of Dipolar Near-Fields.* META 2026 (Dublin, Ireland). July 2026.

P2. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Enabling light-driven 3D control of nanomotors with optical forces.* Researcher's Tuesday (Mons, Belgium). Mar. 2026.

P3. **Guillermo Serrera**, Tanaka, Y. Y. & Albella, P. *Innovative approaches to 3D motion control in plasmonic nanomotors with optical pulling forces.* CEN2025 (Madrid, Spain). June 2025.

P4. **Guillermo Serrera**, González-Colsa, J., González, F., Moreno, F., Saiz, J. M. & Albella, P. *Grating metasurfaces as directional plasmon sources: applications in achiral and chiral sensing.* ImagineNano 2021 - 3PM2021 (Bilbao, Spain). Nov. 2021.

P5. **Guillermo Serrera**, González-Colsa, J., González, F., Moreno, F., Saiz, J. M. & Albella, P. *Enhanced chiroptical activity with dielectric chiral nanostructures: applications in racemic-mixture sensing.* International Conference on Metamaterials and Nanophotonics (Metanano 2021) (Online). Sept. 2021.

Softwares

S1. **Guillermo Serrera**, Gutiérrez, Y. & Gil, J. J. *DiPolIn (DIpolar POLarization in INtrinsic reference frame)* Free software uploaded to Zenodo (DOI: 10.5281/zenodo.18682022). Feb. 2026.

Service

Peer review activity

2026– Journal of the Optical Society of America B

Teaching

More than 170 hours of teaching in the following courses and institutions:

University of Cantabria

2025–26 Teaching Assistant, Physics I (G375)
2024–26 Teaching Assistant, Physics II (G383/G393)
2023–26 Teaching Assistant, Advanced Experimental Techniques (G1775)
2023–24 Teaching Assistant, Energy in the World of Today (G1675)

Teaching-specific training

Beyond teaching activities, 50 hours in teaching-specific training:

2026 Motivation and learning (8 hours), University of Cantabria courses.
2026 Reflection on the own teaching practice in higher education (9 hours), University of Cantabria courses.

2026 The revolution of generative artificial intelligence in research and teaching (12 hours), University of Cantabria courses.

2025 Strategies for Including the SDGs in university teaching (3 hours), University of Cantabria courses.

2023 Creative strategies in teaching and research (6 hours), University of Cantabria courses.

2023 Social competences: tools for university teachers (6 hours), University of Cantabria courses.

2023 Emotional competences: tools for university teachers (6 hours), University of Cantabria courses.

Supervisions

Bachelor theses

2026 Víctor Johan Moreno Cedeño, University of Cantabria, *Polarimetric study of multipolar scattering in arbitrary nanoscatterers.* Co-supervised with Prof. Fernando Moreno.

2024 Bárbara Cacicedo López, University of Cantabria, *Study of structured light scattering by nanostructures: application in optical chirality.* Co-supervised with Prof. Pablo Albella.

Research stays and visits

2025 **University of Fribourg**, two-month research stay at the Photonic Nanosystems Group, supervised by Prof. Guillermo P. Acuña.

2024 **Ghent University**, three-month research stay at the Photonics Research Group, supervised by Prof. Alberto G. Curto.

2021 **Instituto de Estructura de la Materia, CSIC**, visit to Dr. Vincenzo Giannini, attended the seminars "Novel plasmonic metamaterials: applications in sensing", and "Thermal effects in nanoparticles".

2020 **University of Oviedo**, visit to Prof. Pablo Alonso-González, attended the seminar "Advanced near-field optics".

Last updated: October 14, 2026